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Parker weights revisited

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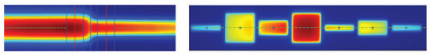

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Letter to the Editor

Parker weights revisited

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To the Editor,

I read with interest the recent article by Wesarg *et al.*[1] regarding the weighting of redundant data in CT reconstruction. Whereas the results of the article are no doubt useful the basic problem and solution may be restated algebraically.

The problem may be considered that of reconstructing an input matrix (of values of linear attenuation coefficient in the case of CT or isotope concentration in the case of PET or SPECT) from a matrix of projection data (the sinogram). These two matrices are in principle related by a linear transformation. If we rearrange the elements of the rectangular input matrix (Ξ) and rectangular sinogram matrix (Λ) into two long column vectors, then we may write

$$\Lambda = W\Xi, \tag{1}$$

where W is the matrix effecting the transformation. Ignoring bandwidth considerations this transformation will in some sense be reversible. Data acquisition in CT or PET is redundant, i.e., there are more elements in Λ than in Ξ and hence W is rectangular. The task at hand is given some experimental determination of Λ (e.g., λ) what is the best estimate of Ξ (e.g., ξ). The least-squares criterion, i.e., minimizing

$$RSS = (\lambda - W\xi)^T(\lambda - W\xi) \tag{2}$$

leads directly to

$$\xi = (W^T W)^{-1} W^T \lambda \tag{3}$$

as the best estimate. The confusion in CT reconstruction with redundant data results from the fact that if the projection sinogram data are perfect, i.e.,

$$\lambda = W\xi, \tag{4}$$

then ξ is obtainable from

$$\xi = (AW)^{-1} A\lambda, \tag{5}$$

where A is any matrix as long as $[AW]$ is nonsingular. A perfect determination of ξ can be obtained from perfect data by a variety of algorithms; one being filtered backprojection. In the presence of noise however, one algorithm may be preferable to another. These points are made by Wesarg *et al.* They also show that discretized data bandwidth requirements limit the ability of filtered backprojection to produce an exact reconstruction. Algebraic reconstruction of images from reconstructions, while computationally intensive, are not subject to the same sampling constraints as filtered backprojection.[2,3]

[1] S. Wesarg, M. Ebert, and T. Bortfeld, ''Parker weights revisited,'' Med. Phys. **29**, 372–378 (2002).
[2] A. H. Anderson, ''Algebraic reconstruction in CT from limited views,'' IEEE Trans. Med. Imaging **8**, 50–55 (1989).
[3] G. H. Glover and N. J. Pelc, ''An algorithm for the reduction of metal clip artifacts in CT reconstructions,'' Med. Phys. **8**, 799–807 (1981).

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